MATHEMATICS N5 (16030186)

PROGRAMME(S):ELECTRICAL,CIVIL,MECHANICAL AND CHEMICAL

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CHAPTER I

LIMITS AND CONTINUITY

1. LEARNING OUTCOMES

Students should be able to evaluate the following:

- Apply L Hospitals rule if a given limit simplifies to an Indeterminate form.
- State the conditions of continuity.
- Detwrmine whether a function is continuous or discontinuous at a certain point or a given interval.
- Calculate points of discontinuity.

ln y
ln y

1 Limits and continuity

1.1 Introduction

We often use graphs and functions to determine values that we cannot measure. Those which we *can* measure, we represent in graphs and express by means of equations. However, we cannot take readings from a graph if we suspect that the graph is not continuous. We must first identify the points of discontinuity because they represent 'breaks' in the graph. If we can 'plug' a 'break' in a graph with a limiting value, we can use this approximate value for all practical purposes.

1.2 Limits

In N3 and N4 mathematics we became conversant with the concept of a 'limit'. In N5 mathematics we consider limits of indeterminate forms and some of the applications of limits in engineering mathematics.

Objective

Once you have completed this section, you should be able to:

Apply L'Hospital's rule if a given limit simplifies to any of the following indeterminate forms:

0^0; $\frac{0}{0}$; $\frac{\infty}{\infty}$; $\infty - \infty$

Preknowledge

The following should refresh your memory on the limits prescribed for N4 mathematics:

1. $\lim\limits_{x \to a} f(x) = f(a)$
2. $\lim\limits_{x \to a} [k.f(x)] = k[\lim\limits_{x \to a} f(x)]$
3. $\lim\limits_{x \to a} [f(x) \pm g(x)] = \lim\limits_{x \to a} f(x) \pm \lim\limits_{x \to a} g(x)$

4. $\lim_{x \to a} f(x)g(x) = \lim_{x \to a} f(x) \lim_{x \to a} g(x)$

5. $\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} f(x) \div \lim_{x \to a} g(x)$

6. $\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)}$

7. $\lim_{x \to \infty} \frac{1}{x} = 0$

8. $\lim_{x \to \infty} \frac{k}{x^n} = 0$

9. $\lim_{x \to \infty} k = k$

As L'Hospital's rule is based on differentiation we also summarise the N4 differentiation formulas for you in this table:

	$y = f(x)$	$\frac{dy}{dx} = f'(x)$
1.	kx^n	nkx^{n-1}
2.	ka^{nx}	$nka^{nx} \ln a$
3.	ke^{nx}	nke^{nx}
4.	$k \ln x$	$\frac{k}{x}$
5.	$k \ln(nx)$	$\frac{k}{x}$
6.	$k \log_a x$	$\frac{k}{x \ln a}$
7.	$k \log_a(nx)$	$\frac{k}{x \ln a}$
8.	$k \sin(nx)$	$nk \cos(nx)$
9.	$k \cos(nx)$	$-nk \sin(nx)$
10.	$k \tan(nx)$	$nk \sec^2(nx)$
11.	$k \cot(nx)$	$-nk \operatorname{cosec}^2(nx)$

	$y = f(x)$	$\frac{dy}{dx} = f'(x)$
12.	$k\sec(nx)$	$nk\sec(nx)\tan(nx)$
13.	$k\,\text{cosec}\,(nx)$	$-nk\,\text{cosec}\,(nx)\cot(nx)$
14.	$f(x)g(x)$ (product)	$f(x)g'(x) + g(x)f'(x)$
15.	$\frac{f(x)}{g(x)}$ (quotient)	$\frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$

1.2.1 Revision of limits of determinate forms

We can use different methods to calculate limits of determinate forms. Some of these methods are illustrated in the examples below.

- **Direct substitution**

The value which the variable approaches, is substituted directly into the function.

Example

Determine: $\lim\limits_{x \to 3} \left(\frac{x^2 - 1}{x^2 + 1}\right)$

Solution:

$$\lim_{x \to 3} \left(\frac{x^2 - 1}{x^2 + 1}\right) = \frac{9 - 1}{9 + 1}$$
$$= 0{,}8$$

- **Simplification**

Try simplification if you obtain an indeterminate form when substituting directly.

Examples

1. Determine: $\lim_{h \to 0} \left(\frac{\frac{1}{x+h} - \frac{1}{x}}{h} \right)$

Solution:

Since we obtain the form $\frac{0}{0}$ after substitution, we simplify the expression further:

$$\therefore \lim_{h \to 0} \left(\frac{\frac{1}{x+h} - \frac{1}{x}}{h} \right) = \lim_{h \to 0} \left[\frac{\frac{x-(x+h)}{x(x+h)}}{h} \right]$$

$$= \lim_{h \to 0} \left[\frac{x - x - h}{x(x+h)} \times \frac{1}{h} \right]$$

$$= \lim_{h \to 0} \left[\frac{-1}{x(x+h)} \right]$$

$$= -\frac{1}{x^2}$$

2. Determine: $\lim_{x \to -1} \left(\frac{x^2 - 1}{x + 1} \right)$

Solution:

If we substitute -1 for x, we obtain the indeterminate form $\frac{0}{0}$. We then manipulate the function to a simpler form by factorising the numerator:

$$\therefore \lim_{x \to -1} \left(\frac{x^2 - 1}{x + 1} \right) = \lim_{x \to -1} \left[\frac{(x-1)(x+1)}{x+1} \right]$$

$$= \lim_{x \to -1} (x - 1)$$

$$= -2$$

▬ Rationalisation

If we obtain a root in a quotient above or below the fraction bar and then obtain an indeterminate form, we simplify by rationalising the numerator or the denominator. We do this by multiplying either the numerator or the denominator by its conjugate.

Examples

1. Determine: $\lim_{x \to 3} \left(\frac{x - 3}{\sqrt{x-2} - \sqrt{4-x}} \right)$

Solution:

Once again, we obtain the form $\frac{0}{0}$, so we rationalise the denominator by multiplying it by its conjugate:

$$\therefore \lim_{x \to 3} \left(\frac{x-3}{\sqrt{x-2} - \sqrt{4-x}} \right)$$

$$= \lim_{x \to 3} \left(\frac{x-3}{\sqrt{x-2} - \sqrt{4-x}} \times \frac{\sqrt{x-2} + \sqrt{4-x}}{\sqrt{x-2} + \sqrt{4-x}} \right)$$

$$= \lim_{x \to 3} \left[\frac{(x-3)(\sqrt{x-2} + \sqrt{4-x})}{(x-2) - (4-x)} \right]$$

$$= \lim_{x \to 3} \left[\frac{(x-3)(\sqrt{x-2} + \sqrt{4-x})}{2x-6} \right]$$

$$= \lim_{x \to 3} \left[\frac{(x-3)(\sqrt{x-2} + \sqrt{4-x})}{2(x-3)} \right]$$

$$= \lim_{x \to 3} \left(\frac{\sqrt{x-2} + \sqrt{4-x}}{2} \right)$$

$$= 1$$

2. Determine: $\lim_{x \to 0} \left(\frac{\sqrt{x+4} - 2}{x} \right)$

Solution:

Once again, we obtain the form $\frac{0}{0}$, so we simplify the expression by rationalising the numerator:

$$\therefore \lim_{x \to 0} \left(\frac{\sqrt{x+4} - 2}{x} \right) = \lim_{x \to 0} \left(\frac{\sqrt{x+4} - 2}{x} \times \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} \right)$$

$$= \lim_{x \to 0} \left[\frac{(x+4) - 4}{x(\sqrt{x+4} + 2)} \right]$$

$$= \lim_{x \to 0} \left(\frac{1}{\sqrt{x+4} + 2} \right)$$

$$= \frac{1}{4}$$

▪ Division by the highest power of x

If in a quotient the variable tends to ∞, and we obtain the indeterminate form $\frac{\infty}{\infty}$, we divide both the numerator and the denominator by the highest power of x occurring in the quotient. We then make use of the fact that $\frac{1}{\infty} = 0$ or, in general, $\frac{c}{\infty} = 0$.

Example

Determine: $\lim\limits_{x \to \infty} \left(\frac{2 - 2x + x^3}{x - 2x^2 + 5x^3} \right)$

Solution:

Since we obtain the form $\frac{\infty}{\infty}$, we divide above and below the fraction bar by the highest power of x, which is x^3.

$$\therefore \lim_{x \to \infty} \left(\frac{2 - 2x + x^3}{x - 2x^2 + 5x^3} \right) = \lim_{x \to \infty} \left(\frac{\frac{2}{x^3} - \frac{2}{x^2} + 1}{\frac{1}{x^2} - \frac{2}{x} + 5} \right)$$

$$= \frac{1}{5}$$

1.2.2 L'Hospital's rule

In all the above examples it was possible to manipulate the function to a determinate form. This, however, does not hold for each and every indeterminate form, as in the case of $\lim\limits_{x \to 0} \left(\frac{\tan x - x^2}{x^2 - \sin x} \right) = \frac{0}{0}$

We cannot manipulate this form to a determinate form, hence we use L'Hospital's rule, a more general method of calculating the limits of indeterminate forms such as $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

L'Hospital's rule:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)} = \lim_{x \to a} \frac{f''(x)}{g''(x)} = \ldots \text{ etc.}$$

Remember:

L'Hospital's rule is applicable only to fractions. Therefore, if we obtain an indeterminate form such as 0° or $\infty - \infty$, we must first manipulate it to obtain a fraction or a quotient.

Examples

Determine the following limits:

1. $\lim_{x \to -3} \left(\frac{x^3 + 27}{x^2 - 9} \right)$

Solution:

$$\lim_{x \to -3} \left(\frac{x^3 + 27}{x^2 - 9} \right) = \frac{0}{0}$$

Apply L'Hospital's rule by differentiating the numerator and the denominator:

$$\therefore \lim_{x \to -3} \left(\frac{x^3 + 27}{x^2 - 9} \right) = \lim_{x \to -3} \left(\frac{3x^2}{2x} \right)$$

$$= \frac{27}{-6}$$

$$= -4{,}5$$

> **Remember:**
> Always show the indeterminate form and state your intention to use L'Hospital's rule.

2. $\lim_{x \to \infty} \left(\frac{3x^3}{e^{3x}} \right)$

Solution:

$$\lim_{x \to \infty} \left(\frac{3x^3}{e^{3x}} \right) = \frac{\infty}{\infty}$$ Use L'Hospital.

$$\therefore \lim_{x \to \infty} \left(\frac{3x^3}{e^{3x}} \right) = \lim_{x \to \infty} \left(\frac{9x^2}{3e^{3x}} \right)$$

$$= \lim_{x \to \infty} \left(\frac{3x^2}{e^{3x}} \right)$$ Type $\frac{\infty}{\infty}$.

$$= \lim_{x \to \infty} \left(\frac{6x}{3e^{3x}} \right)$$

$$= \lim_{x \to \infty} \left(\frac{2x}{e^{3x}} \right)$$ Type $\frac{\infty}{\infty}$.

$$= \lim_{x \to \infty} \left(\frac{2}{3e^{3x}} \right)$$

$$= 0$$

3. $\lim_{x \to 0} \left(\frac{\ln x}{\cot x} \right)$

Solution:

$$\lim_{x \to 0} \left(\frac{\ln x}{\cot x} \right) = \frac{\infty}{\infty}$$ Use L'Hospital.

$$\therefore \lim_{x \to 0} \left(\frac{\ln x}{\cot x} \right) = \lim_{x \to 0} \left(\frac{\frac{1}{x}}{-\operatorname{cosec}^2 x} \right)$$ Type $\frac{\infty}{\infty}$.

$$= \lim_{x \to 0} \left(\frac{-\sin^2 x}{x} \right)$$ Type $\frac{0}{0}$.

$$= \lim_{x \to 0} \left(\frac{-2 \sin x . \cos x}{1} \right)$$

$$= \frac{0}{1}$$

$$= 0$$

4. $\lim_{x \to 1} \left(\frac{x}{\ln x} - \frac{1}{\ln x} \right)$

Solution:

$$\lim_{x \to 1} \left(\frac{x}{\ln x} - \frac{1}{\ln x} \right) = \infty - \infty$$

As this is not a quotient, manipulate this form to obtain a fraction first. Then use L'Hospital.

$$\therefore \lim_{x \to 1} \left(\frac{x}{\ln x} - \frac{1}{\ln x} \right) = \lim_{x \to 1} \left(\frac{x - 1}{\ln x} \right)$$ Type $\frac{0}{0}$.

$$= \lim_{x \to 1} \left(\frac{1}{\frac{1}{x}} \right)$$

$$= 1$$

Remember:

The only two trigonometric limits that are defined, are:

1. $\lim_{x \to a} \sin x = \sin a$
2. $\lim_{x \to a} \cos x = \cos a$

Note: The only indeterminate form that we have not dealt with, is 0^0. This form is dealt with in the following example.

Example

Determine: $\lim_{x \to 0} (\sin x)^{\tan x}$

Solution:

$\lim_{x \to 0} (\sin x)^{\tan x} = 0^0$

Change to a fraction.

Let $y = (\sin x)^{\tan x}$

then $\ln y = \tan x . \ln(\sin x)$

$= \dfrac{\ln(\sin x)}{\cot x}$

Type $\dfrac{\infty}{\infty}$.

$\therefore \lim_{x \to 0} \ln y = \lim_{x \to 0} \left(\dfrac{\ln(\sin x)}{\cot x} \right)$

Use L'Hospital.

$= \lim_{x \to 0} \left(\dfrac{\cot x}{-\operatorname{cosec}^2 x} \right)$

$= \lim_{x \to 0} -(\sin x . \cos x)$

$= 0$

$\therefore \lim_{x \to 0} y = 1$

Remember:

$\ln y = 0$

$y = e^0$

$y = 1$

Exercise 1.1

Determine the following limits by using L'Hospital's rule:

1. $\lim_{x \to -1} \left(\dfrac{x^5 + 1}{x^3 + 1} \right)$
2. $\lim_{x \to 3} \left(\dfrac{\sqrt{x} - \sqrt{3}}{x - 3} \right)$
3. $\lim_{x \to 0} \left(\dfrac{\sec x - 1}{\cos x - 1} \right)$
4. $\lim_{x \to 0} \left(\dfrac{x + e^x - 1}{xe^x + x} \right)$
5. $\lim_{x \to \infty} \left(\dfrac{2x^2}{e^x} \right)$
6. $\lim_{x \to \infty} \left(\dfrac{x^2 - 1}{3 - x} \right)$
7. $\lim_{x \to 0} \left(\dfrac{\tan 3x}{\tan x} \right)$
8. $\lim_{x \to 0} \left(\dfrac{\ln x}{x^{-1}} \right)$
9. $\lim_{x \to 1} \left(\dfrac{x}{x - 1} - \dfrac{1}{\ln x} \right)$
10. $\lim_{x \to 0} \left(\dfrac{\cos 2x}{x^2} - \dfrac{1}{x^2} \right)$
11. $\lim_{x \to 0} \left(\dfrac{\tan x}{x} \right)$
12. $\lim_{x \to 0} \left(\dfrac{x}{\tan x} \right)$